Run report: `koblitz-algebraic` on koblitz a=0

Observations only. A smoke run of the sweeper on a small space; it is not a result about index calculus.

What ran

- Command: `target/release/ic --json sweep run --space koblitz-algebraic --koblitz-a 0 --rungs 13,15,19,23 --target-log2r 81 --trials 24 --workers 3 --repeats 1 --rho-runs 2 --max-trials 100000 --startup-trials 6 --prune-startup 3 --timeout-seconds 120 --max-rss-mb 4096 --store store.jsonl`
- Commit: not recorded (the build could not read it from a git worktree; see the Host line); binary BLAKE3 `dcdae5089963ec6093fc558913f5daeb7b20427504acf0b9a7e7a303e356cd05`
- Host: Apple silicon (arm64) macOS, shared with other work; the binary was built from commit a0173bca92 plus the then-uncommitted solver-unit guard, committed afterwards as a5c51d3757. Isolation level: L0 (shared host, no CPU pinning, 3 trials at a time). Counts are unaffected; wall time is not used by the score.
- Track: **algebraic** (counted). Rungs requested: [13, 15, 19, 23]; usable (baseline scored): [13, 15, 19, 23].
- Target for the projection: log2 r = 81.0; `min_fit` 4; repeats 1; seed 1; counted-trial cap 100000.
- Raw records: `store.jsonl` (every baseline and trial line, as written).

Baseline (frozen before the search)

rung n	class	S(w=0)	S(w=0.1)	S(w=1)	detail
13	ok	18.25	166.9	1505	
15	ok	33.28	928.4	8985	
19	ok	14.54	3516	3.503e+04	
23	ok	13.14	4.653e+04	4.652e+05	

Trials

Ratio columns are log2 of the worst-case (over w) `S_trial / S_baseline`; negative would be better than the baseline.

id	state	solver	m	dim_frac	n=13	n=15	n=19	n=23	objective	note
1	failed	crossbred-f2	2	0.18	—				—	rung 13: not rankable: solver counts in word operations (partial: ext
4	failed	satcdcl	3	0.16	—				—	rung 13: exhausted — 0 of 1 repeats recovered the logawithin
2	failed	bergerfold-f2	2	0.24	+2.30	+1.87	+3.00	+4.70	+15.47	exhausted 2

How the trials ended

outcome	trials
pruned	8
complete	6
failed: ok: not rankable: solver counts in `wo | 3 | | failed: exhausted: wall-clock limit of 120 s reached | 3 | | failed: exhausted: 0 of 1 repeats recovered the logar | 2 | | failed: ok: not rankable: solver counts inco`	1
failed: ok: not rankable: solver counts in `mo	1

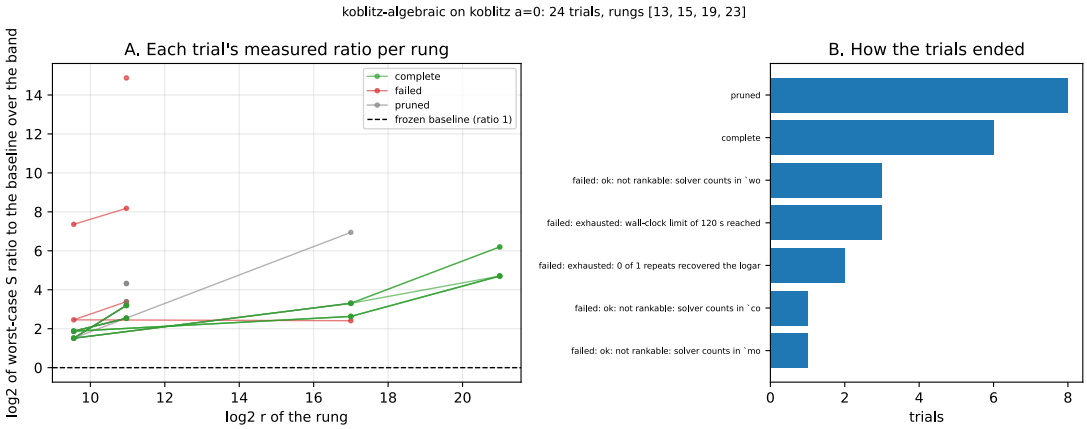


Figure 1: ratios by rung

Figure: recorded values from `store.jsonl`; no value is computed in the plot. Left: each trial's log2 worst-case ratio at the rungs it completed. Right: trial outcomes.

Limits

- Few trials on a small space; this says nothing about which configuration is best.
- Rungs below `min_fit` are not projected: the objective is then the last rung's measured ratio.
- K_0 at n = 15 has a larger cofactor than the cofactor-4 rungs, so the rungs mix cofactors.
- Solver rows in a different unit than the baseline's are recorded and not rankable; see the failed rows' notes.